

java ==> HLL ==> English Words + Symbols(Operators)

operators are symbols

whenever we are cretaining variables

1) variables names should be meaningful
int price(valid), int a(invalid)

2) variable names should always be camel case

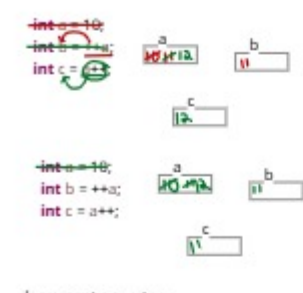
Arithmetic Operators

+ ==> addition
- ==> sub
* ==> mul
/ ==> div
% ==> modulo

Increment Operator

++ ==> 1.0

will always increase ethe value by +1



decrement operator

--

++a ==> pre increment

a++ ==> post increment

--a ==> pre decrement

a-- ==> post decrement

conditional operator

<, >, <=, >=, ==, !=

Logical Operator

logical and ==> &&

logical or ==> ||

logical not ==> !

Assignment Operator

=, +=, -=, *=, /=, %=

Ternary Operator ==> ?

will reduce the length of the program

written usign if - else

1. A shopping application needs to calculate the total cost of 6 notebooks,

where each notebook costs ₹45.

Which operator should be used to calculate the total?

- A) /
- B) %
- C) *
- D) -

2. A banking application needs to

find how much money remains after paying a bill from a

wallet balance. Which operation is appropriate?

- A) Addition
- B) Subtraction
- C) Multiplication
- D) Division

3. A school application needs to distribute 100

chocolates equally among 8 students and determine

how many chocolates are left over.

Which operator is required to find the leftover chocolates?

- A) /
- B) %
- C) *
- D) +

4. A programmer wants to store the number of

books in a variable. The variable name should

contain two words: number and books.

Which name follows the usual java naming convention?

- A) NumberBooks
- B) number_books
- C) numberBooks
- D) NUMBERBOOKS

5. A game increases a player's score by exactly

1 whenever the player collects a coin.

Which operator is most suitable?

- A) --
- B) ++
- C) += 2
- D) %=

6. Consider the following code:

```
int count = 7;
```

```
int result = ++count;
```

```
System.out.println(count);
```

```
System.out.println(result);
```

What is the output?

- A) 78
- B) 8
- C) 7
- D) 8
- E) 8
- F) 7

7. Consider:

```
int tickets = 15;
```

```
int sold = tickets++;
```

```
System.out.println(tickets);
```

```
System.out.println(sold);
```

What is the output?

- A) 15
- B) 16
- C) 16
- D) 15
- E) 16
- F) 15

8. A program keeps track of the number of

active users. When one user logs out, the

count must decrease by exactly 1.

Which operator is appropriate?

- A) ++
- B) --
- C) +=
- D) *=

9. Consider:

```
int x = 20;
```

```
int y = --x;
```

```
System.out.println(x);
```

```
System.out.println(y);
```

What is printed?

- A) 20
- B) 19
- C) 19
- D) 20
- E) 19
- F) 20

10. A billing program has a variable containing the current

bill amount. The programmer wants to increase it

by ₹500 without writing the variable name twice.

Which statement is appropriate?

- A) bill += 500;
- B) bill ++ 500;
- C) bill ++ 500;
- D) bill = 500;

11. A student's marks need to be reduced by 5 because of

a penalty. Which statement correctly performs this

operation?

- A) marks += 5;
- B) marks -= 5;
- C) marks *= 5;
- D) marks /= 5;

12. A company doubles the salary stored in a variable

as part of a calculation. Which statement performs

this operation?

- A) salary += 2;
- B) salary -= 2;
- C) salary *= 2;
- D) salary /= 2;

13. A program needs to reduce a value to one-fourth of

its current value. Which statement is appropriate?

- A) value *= 4;
- B) value += 4;
- C) value -= 4;
- D) value /= 4;

14. Consider:

```
int points = 13;
```

```
points %= 5;
```

```
System.out.println(points);
```

What will be printed?

- A) 2
- B) 3
- C) 5
- D) 8

15. A program needs to check whether a

customer's age is at least 18 before

allowing registration.

Which operator should be used?

- A) <
- B) >
- C) >=
- D) !=

16. A temperature monitoring application needs to

determine whether the temperature is below 0°C.

Which operator would be appropriate?

- A) >
- B) <
- C) >=
- D) ==

17. A login system wants to check whether the entered

PIN exactly matches the stored PIN.

Which operator should be used?

- A) =
- B) ==
- C) !=
- D) >=

18. A validation program wants to make sure that

two usernames are not the same.

Which operator is appropriate?

- A) ==
- B) =
- C) !=
- D) <=

19. A program should allow a discount only

when the customer is a premium member **and**

has purchased at least 10 items.

Which logical operator is required to

combine these two conditions?

- A) ||
- B) !
- C) &&
- D) %

20. A transport application should allow a passenger

to use either a student ID **or** a senior citizen ID to

receive a concession. Which logical operator is appropriate?

- A) &&
- B) ||
- C) !
- D) ==

21. A security system should allow access when a user is **not** blocked. Which logical

operator can reverse the Boolean condition?

- A) &&
- B) ||
- C) !
- D) %

22. Consider:

```
int age = 25;
```

```
boolean hasId = true;
```

```
if (age >= 18 && hasId) {
```

```
    System.out.println("Allowed");
```

```
}
```

What will be printed?

- A) Denied
- B) Allowed
- C) true
- D) Nothing

23. Consider:

```
boolean isHoliday = false;
```

```
boolean isWeekend = true;
```

```
if (isHoliday || isWeekend) {
```

```
    System.out.println("No work");
```

```
}
```

What will be printed?

- A) Work
- B) No work
- C) false
- D) Nothing

24. Consider:

```
boolean available = false;
```

```
if (!available) {
```

```
    System.out.println("Item unavailable");
```

```
}
```

What will be printed?

- A) Item available
- B) false
- C) Item unavailable
- D) Nothing

25. A program needs to read a whole

number entered by a user,

such as the number of passengers.

Which Scanner method should be used?

- A) nextDouble()
- B) nextFloat()
- C) nextInt()
- D) nextBoolean()

26. A program asks the user to enter the weight of a

package, such as 12.75. Which method is suitable for

reading the value as a double?

- A) nextInt()
- B) nextDouble()
- C) nextLine()
- D) nextBoolean()

27. A program asks the user to enter a decimal

measurement and wants to store it in a float variable.

Which method should be used?

- A) nextInt()
- B) nextFloat()
- C) nextDouble()
- D) nextBoolean()

28. A movie application wants to display "Adult" when a person's

age is 18 or above and "Minor" otherwise.

The programmer wants to accomplish this using a

single expression rather than a complete if-else statement.

Which Java feature is appropriate?

- A) Modulo operator
- B) Ternary operator
- C) Increment operator
- D) Assignment operator

29. Consider:

```
int marks = 72;
```

```
String result = marks >= 40 ? "Pass" : "Fail";
```

```
System.out.println(result);
```

What will be printed?

- A) 72
- B) true
- C) Pass
- D) Fail

30. A developer is creating a java application and needs to read input from the keyboard.

Which statement correctly creates the object required for keyboard input?

- A) Scanner input = Scanner();
- B) Scanner input = new Scanner(System.in);
- C) Scanner = new input(System.in);
- D) new Scanner = input(System.in);